Newton’s Laws

Building Models to Describe our World

Newton, Forces, and Free Body Diagrams

Outline

Concepts

Building Models to Describe our World

Newton, Forces, and Free Body Diagrams

What is a force?

- There is no such thing as a force.
- These are a mathematical tool that allows us to describe the motion of objects.
- We represent them with a vector, as they have a magnitude and a direction.
- We can use a spring scale to define the magnitude and direction of a force.
- One issue is that we have some intuition about what a force feels like; it's better to apply rules for modelling forces, rather than trying to guess and apply intuition. Some concepts are counter-intuitive.
    - With practice, you will develop intuition about whether to include a specific force, not about what a force really is. The "physics intuition" is about getting comfortable applying the rules, not about understanding them. Nobody "understands" the rules.

Newton’s first law

An object will remain in its state of motion, be it at rest or moving with constant velocity, unless a net external force is exerted on the object.

If the above is true, then you’re observing the object from an inertial frame of reference. Think of this as the way to define an inertial frame of reference.

Newton's second law

$$\sum \vec{F} = m\vec{a}$$

- This statement is a quantitative framework that describes almost all of the world around us. It leads to the concepts of conservation of energy, momentum, angular momentum, etc. It is an amazing equation!
- It connects dynamics (forces) to kinematics (acceleration).
- It is a vector equation $\rightarrow$ To be "useful" we need to write it out in components.

Newton’s third law

If one object exerts a force on another object, the second object exerts a force on the first object that is equal in magnitude and opposite in direction.

This is not about intuition!!! This is a rule to be followed! (science)

If you include a force on object A that is exerted by object B, then you need to include an equal and opposite force exerted on object B from object A (if you care about modeling the motion of object B).

Forces

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Identifying forces

- It’s important, it’s important, it’s important
- You want to be certain that you did not miss any force exerted on an object, or your model is almost definitely wrong!
- Ask yourself: “What could be pulling or pushing on the object?” and go through a list of possible forces.

Identifying the forces – multiple objects

- Don’t forget Newton’s third law!
  - If you have multiple objects make sure that any force exerted by one object on the other has a matching “reaction” force.
- It’s usually easiest to start with the object that has the least number of forces exerted on it (for example, the “top” block, if there are stacked blocks).

Types of forces (not an exhaustive list, but covers most cases)

- **Weight**: Is the object near the surface of a planet?
- **Normal force**: Is the object in contact with anything else?
- **Frictional forces**: Can there be friction associated with a normal force?
- **Tension forces**: Is something pulling on the object, maybe with a rope?
- **Drag forces**: Is the object moving through a fluid?
- **Spring forces**: Is the object in contact with a spring?
- **Inertial forces**: Are you in a non-inertial frame of reference?
- **"Applied forces"**: Is anything else, like a human, pushing or pulling on the object?

Friction

- Force between two surfaces, in the direction parallel to the interface between surface (perpendicular to the corresponding normal force).
- **Only where there is a normal force** between two surfaces.
- **Static friction**:
  - When the surfaces are not moving relative to each other.
  - Can have a maximal value:

    $$f_s^{max} = \mu_s N$$

    but the magnitude depends on the other forces (typically, the sum of the forces must be zero, since static friction usually implies no acceleration).
  - Direction is always opposite to that of “impeding motion”.
- **Kinetic friction**:
  - When surfaces are moving/sliding relative to each other.
  - Magnitude depends on the normal force:

    $$f_k = \mu_k N$$

  - Direction is always opposite to that of the motion (acts to decelerate).

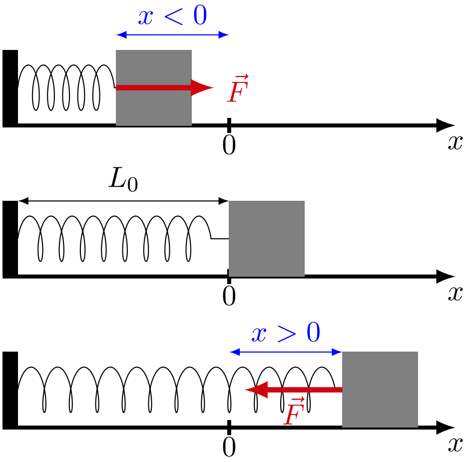
Hooke's law

- Springs have a rest length, and a "spring constant" (how stiff they are), k.
- If you compress them or stretch them by a displacement x relative to their rest position, they will exert a force that is proportional to x:

$$F = kx \quad \text{(magnitude)}$$

- That spring force (or "restoring force" will be in the direction opposite of the displacement:

$$\vec{F} = -kx\hat{x} \quad \text{(magnitude and direction)}$$



A spring-mass system. When the moving end of the spring is at $x = 0$, there is no spring force. When the moving end is in negative x (compression), the force is in the positive direction. When the moving end is in positive x (extension), the force is in the negative direction.

Free Body Diagrams

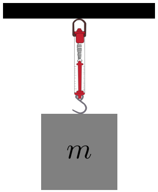
Building Models to Describe our World

Newton, Forces, and Free Body Diagrams

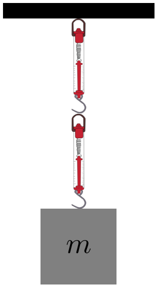
Using Newton's second law

1. Identify the forces (it's kinda important).
2. Make a *good* FBD for each object with a mass:
   2.1 Show the forces.
   2.2 Show the acceleration vector.
   2.3 Show your choice of coordinate system (a good choice is $\hat{x}$ parallel to $\vec{a}$ for that mass).
   2.4 Show any relevant angles.
3. Use the FBD to write out the components of Newton's Second Law for each mass.

Group exercise: Draw FBD for each object



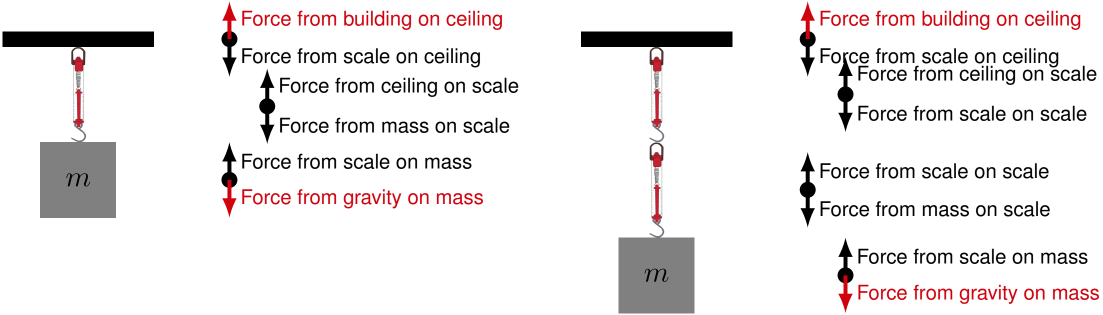
- ceiling
  - scale
- mass



- ceiling
  - scale
- scale
  - mass

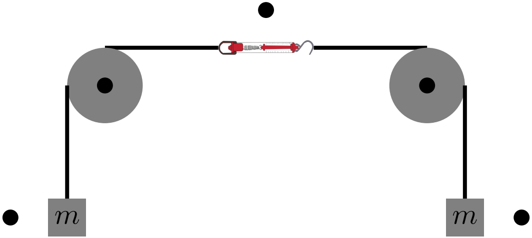
- Assume the scales are massless.
- Draw free body diagrams for the masses, scales and ceiling in both cases. (Draw little arrows around each point!).
- Remember, all objects must have an acceleration of zero!
- Be clear about what exerts a force on what!

Group exercise: Draw FBD for each object



- Two forces on each object, and they must have same magnitude/opposite direction.
- The Earth and building (which holds up the ceiling) exert forces but are not shown. We only show bodies on which forces are exerted.
- You need to become comfortable identifying forces that are exerted on objects!

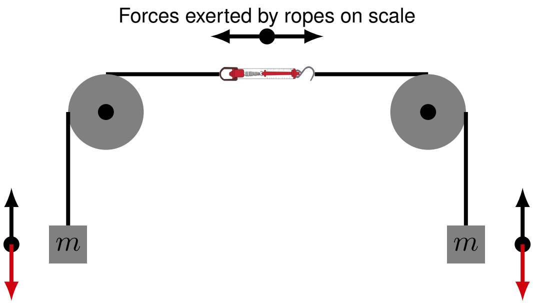
Group exercise: Draw the FBDs for masses and scale



Two masses, two pulleys and a spring scale.

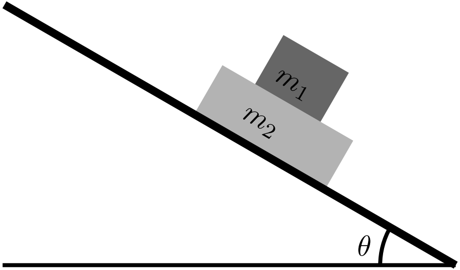
Group exercise: Draw the FBDs for masses and scale

- All forces have the same magnitude, but the pulleys change the direction of tension forces in the ropes.
- The forces exerted on the scale are the same as in the previous problem, so it reads the same!
- The Earth exerts the two forces of gravity (in red).



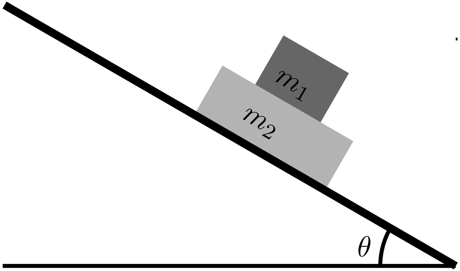
Two masses, two pulleys and a spring scale

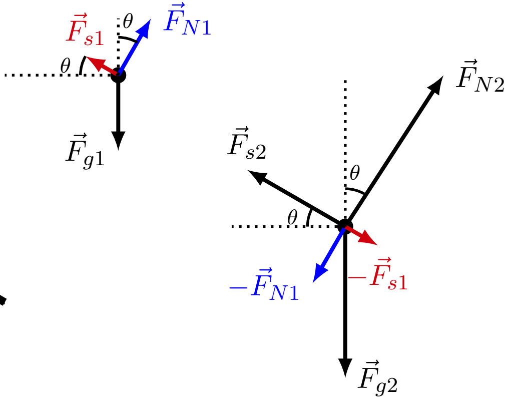
Draw the FBD for each block



Two blocks on an inclined plane.

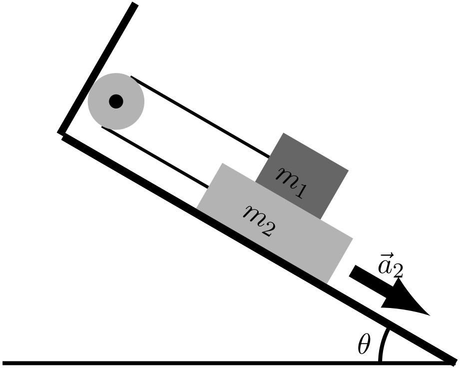
Draw the FBD for each block





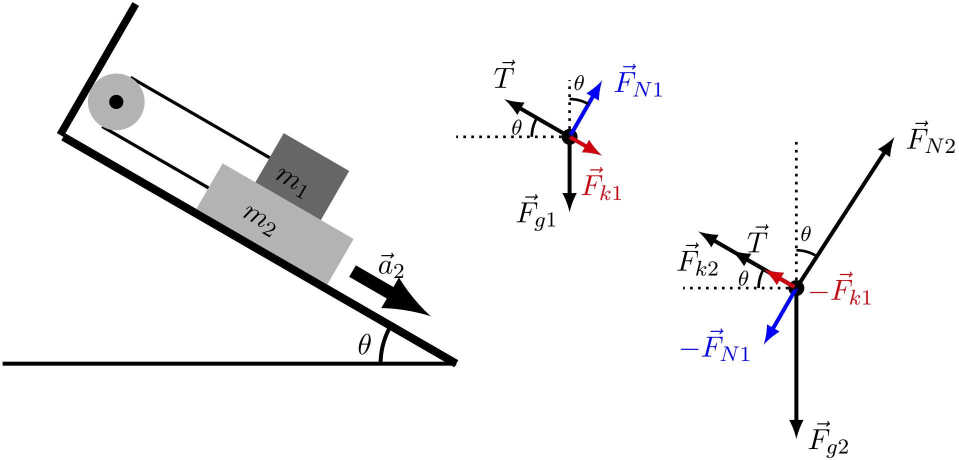
Two blocks on an inclined plane, with free body diagram. Arrows of the same colour represent action and reaction "pairs", which are necessarily exerted on different bodies (Newton's Third Law).

Draw the FBD for each block



Two blocks on an inclined plane with pulley.

Draw the FBD for each block



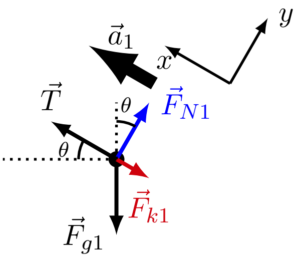
Two blocks on an inclined plane with pulley, with free body diagram. Arrows of the same colour represent action and reaction "pairs", which are necessarily exerted on different bodies (Newton's Third Law).

Applying Newton's Second Law to block 1

We "simply" read off the FBD to obtain the two components of Newton's Second Law:

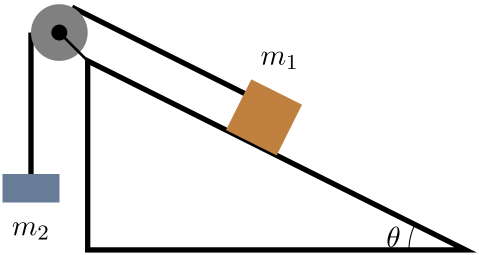
$$\begin{aligned}\sum F_x \rightarrow T - F_{k1} - F_{g1}\sin\theta &= m_1 a_1 \\ \sum F_y \rightarrow F_{N1} - F_{g1}\cos\theta &= 0\end{aligned}$$

You would do the same for block 2, noting that $a_1 = a_2$, since the *magnitude* of the acceleration of the two blocks must be the same (they are connected by a rope).



Free-body diagram for m_1 from the previous slide.

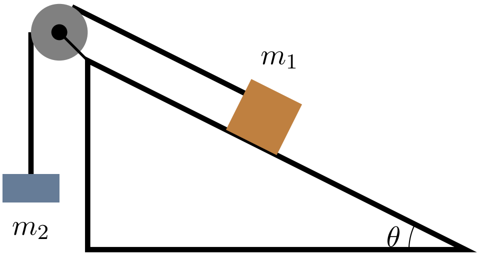
Blocks on an incline



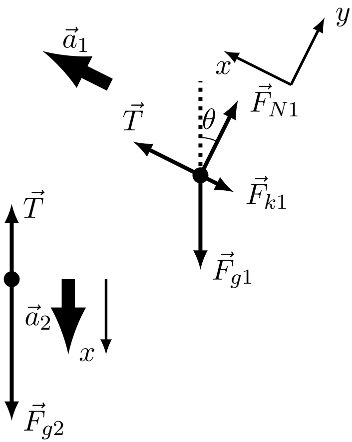
Two blocks connected by a pulley.

Draw the free body diagrams for both block when they accelerate.

Blocks on an incline



Two blocks connected by a pulley.



Free body diagrams for the two blocks.